

Experiment No. 2

Test Case 1

Analyze the SWR of a transmission line for a fixed characteristic impedance (Z_0) of 50 ohms, as the load impedance (Z_L) is varied across 25 ohms to 100 ohms. Verify how the SWR changes with each value of Z_L . When $Z_L = Z_0$, the SWR should equal 1 (indicating no reflection).

PROGRAM:

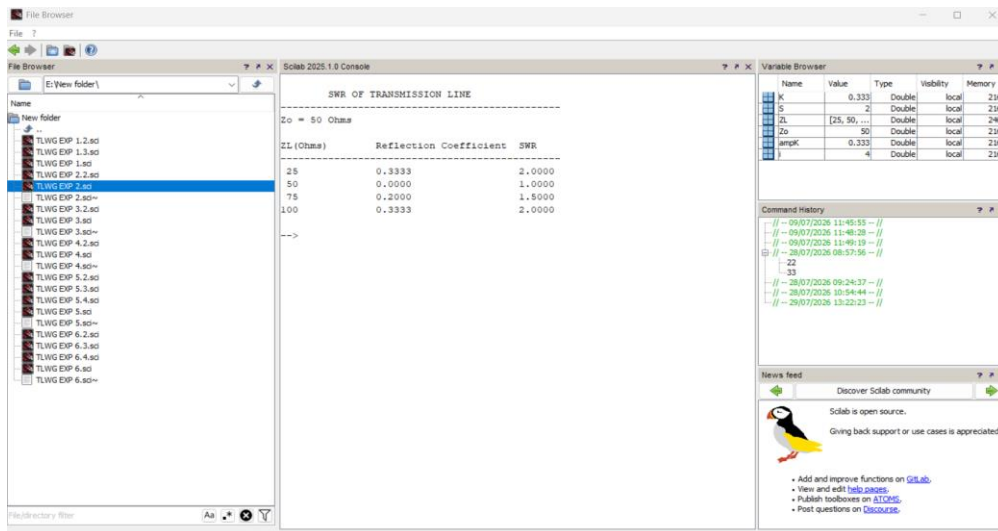
```
clc; clear; close;
Zo=50; ZL=[25,50,75,100];
for i=1:4
    K=(ZL(i)-Zo)/(ZL(i)+Zo);
    ampK=abs(K); S=(1+ampK)/(1-ampK);
    printf("ZL=%f   |K|=%f   SWR=%f\n",ZL(i),ampK,S);
end
```

OUTPUT:

Zo = 50 ohms — Variable: ZL

ZL (Ω)	K — Simulation	SWR — Simulation	K — Theoretical	SWR — Theoretical
25	0.3333	2.0000	0.3333	2.0000
50	0.0000	1.0000	0.0000	1.0000
75	0.2000	1.5000	0.2000	1.5000
100	0.3333	2.0000	0.3333	2.0000

OUTPUT SCREENSHOT:



Test Case 2

Analyze the SWR for a fixed load impedance (ZL) of 50 ohms while varying the characteristic impedance (Zo) of the transmission line from 50 ohms to 400 ohms. Verify how the SWR changes with each value of Zo, including Zo = ZL.

PROGRAM:

```
clc; clear; close;
ZL=50; Zo=[50,250,300,350,400];
for i=1:5
    K=(ZL-Zo(i))/(ZL+Zo(i));
    ampK=abs(K); S=(1+ampK)/(1-ampK);
    printf("Zo=%f   |K|=%f   SWR=%f\n",Zo(i),ampK,S);
end
```

OUTPUT:

Fixed Load Impedance ZL = 50 ohms — Variable: Zo

Zo (Ω)	K — Simulation	SWR — Simulation	K — Theoretical	SWR — Theoretical
50	0.0000	1.0000	0.0000	1.0000
250	0.6667	5.0000	0.6667	5.0000
300	0.7143	6.0000	0.7143	6.0000
350	0.7500	7.0000	0.7500	7.0000
400	0.7778	8.0000	0.7778	8.0000

OUTPUT SCREENSHOT:

The screenshot shows the MATLAB R2020b environment. The Command Window displays the following output:

```
SWR of TRANSMISSION LINE
Fixed Load Impedance (ZL) = 50 Ohms
Zo (Ohms)   Reflection Coefficient   SWR
-----
50           0.0000                    1.0000
250          0.6667                    5.0000
300          0.7143                    6.0000
350          0.7500                    7.0000
400          0.7778                    8.0000
```

The Variable Browser shows the following variables:

Name	Value	Type	Visibility	Memory
K	-0.778	Double	local	236 B
SWR	8	Double	local	236 B
ZL	50	Double	local	236 B
Zo	1x5	Double	local	248 B
i	5	Double	local	236 B
rho	0.778	Double	local	236 B

The Command History shows the following commands:

```
// -- 09/07/2020 11:45:53 -- //
// -- 09/07/2020 11:46:28 -- //
// -- 09/07/2020 11:46:18 -- //
// -- 28/07/2020 08:57:56 -- //
22
33
// -- 28/07/2020 09:24:37 -- //
// -- 28/07/2020 10:54:44 -- //
// -- 29/07/2020 13:22:23 -- //
```